Misconceptions

Misconception 1

Torsional shear produces shear stresses on the cross section as shown, thus, the shear stresses should distort the cross section.

Fact:

No. The cross section is not distorted.

Click here to check the Animation

Misconception 2

If the bar is twisted, its length also changes.

Fact:

No. The length of the bar is unaffected.

Click here to check the Animation

Misconception 3

If there is a small slit made along the bar shown. Since all the properties remain almost the same, the bar will twist to the same extent.

Fact:

No.

Click here to check the Animation

Misconception 4

If a composite rod is subjected to torque T, each of the composite is inturn subjected to torsional load T.

Fact:

No.

Click here to check the Animation

Misconception 5

The total angle of the twist of the bar is same as the shear strain.

Fact:

No.

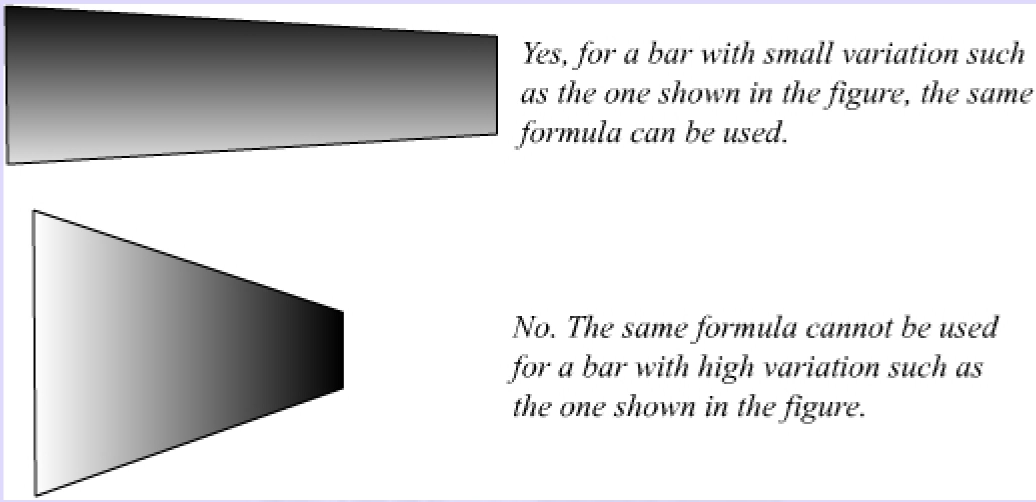
Click here to check the Animation

Misconception 6

For a bar of varying cross section, the same formula can be used!

Fact:

Yes, but for a bar with small variation and not for a bar with a steep variation.



Misconception 7

The bar becomes thinner as you twist it!

Fact:

No. From **assumption2**, we see that the cross section remains the same after twisting.

Misconception 8

The shear stress is maximum at the skin but it is a free surface. Therefore, should'nt the shear stress be zero?

Fact:

No.